NPTEL ONLINE CERTIFICATION COURSES




Course Name: Software Project Management

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Department : Computer Sc. & Engg. NIT Rourkela

TOPIC: Project Monitoring and Control




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## CONCEPTS COVERED

- Code Review
- Code Walk Through
- Code Inspection
- Cleanroom Technique



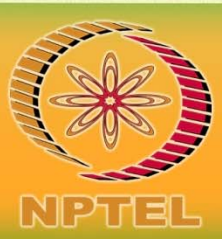

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# Candidate work products for review

- All interim and final work products are usually candidates for review.
- Usually, the work products considered to be suitable candidates for review are as follows.
  - Requirements specification documents
  - User interface specification and design documents
  - Architectural, high-level, and detailed design documents
  - Source code
  - Test plan and the designed test cases
  - Project management plan and configuration management plan

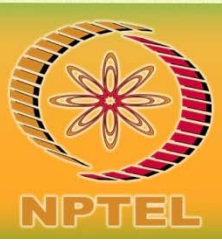


# Code Review

- Review is a very effective technique to remove defects from source code.
- In fact, review has been acknowledged to be more cost-effective in removing defects as compared to testing.
- Over the years, review techniques have become extremely popular and have been generalised for use with other work products.

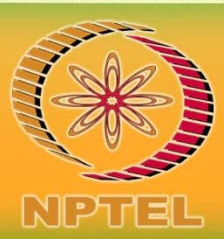


# Code Review cont...

- Code review for a module is undertaken after the module successfully compiles, i.e. all the syntax errors have been eliminated from the module.
- Obviously, code review does not target to design syntax errors in a program, but is designed to detect logical, algorithmic, and programming errors.
- Code review has been recognised as an extremely cost-effective strategy for eliminating coding errors and for producing high quality code.

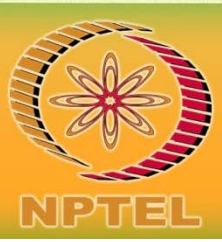


# Code Review cont...

- The reason behind why code review is a much more cost-effective strategy to eliminate errors from code compared to testing is that *reviews directly detect errors*.
- On the other hand, testing only helps detect failures and significant effort is needed to locate the error during debugging.

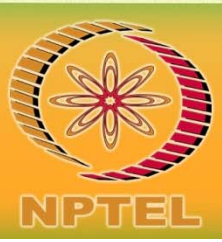


# Code Review cont...

- The rationale behind the above statement is explained as follows. Eliminating an error from code involves three main activities—testing, debugging, and then correcting the errors.
- Testing is carried out to detect if the system fails to work satisfactorily for certain types of inputs and under certain circumstances. Once a failure is detected, debugging is carried out to locate the error that is causing the failure and to remove it.

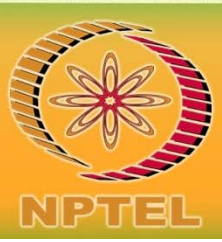


# Code Review cont...

- Of the three testing activities, debugging is possibly the most laborious and time consuming activity.
- In code inspection (which is a type of review), errors are directly detected, thereby saving the significant effort that would have been required to locate the error.



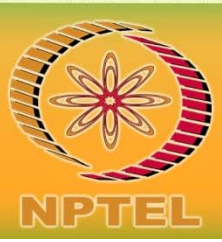



# Code Review cont...

Normally, the following two types of reviews are carried out on the code of a module:

- Code walkthrough.
- Code inspection.

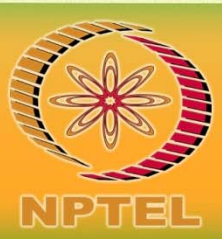


# Code inspection and code walk through

- After a module has been coded,
  - code inspection and code walk through are carried out
  - ensures that coding standards are followed
  - help detect as many errors as possible before testing.

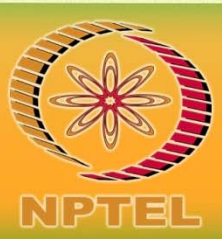


# Code inspection and code walk through cont...

- Detect as many errors as possible during inspection and walkthrough:
  - detected errors require less effort for correction
    - much higher effort needed if errors were to be detected during integration or system testing.

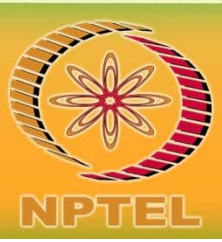


# Code Walk Through

- An informal code analysis technique.
  - undertaken after the coding of a module is complete.
- A few members of the development team select some test cases:
  - simulate execution of the code by hand using these test cases, i.e. the members mentally trace the execution through different statements and functions of the code.

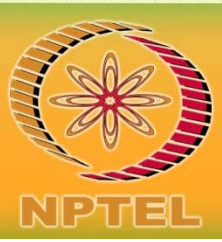


# Code Walk Through cont...

- The members note down their findings of their code walk through and discuss those in a walkthrough meeting.
- Even though it is an informal technique:
  - several guidelines have evolved over the years
  - making this naive but useful analysis technique more effective.
  - These guidelines are based on
    - personal experience, common sense, and several subjective factors.

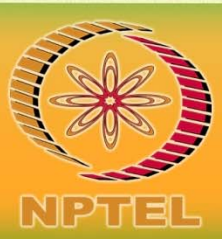


# Code Walk Through cont...

- The guidelines should be considered as examples:
  - rather than accepted as rules to be applied dogmatically.
- The team performing code walk through should not be either too big or too small.
  - Ideally, it should consist of between three to seven members.

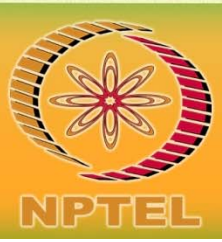


# Code Walk Through cont...

- Discussion should focus on discovery of errors:
  - and not on how to fix the discovered errors.
- To foster cooperation:
  - avoid the feeling among engineers that they are being evaluated in the code walk through meeting,
  - managers should not attend the walk through meetings.

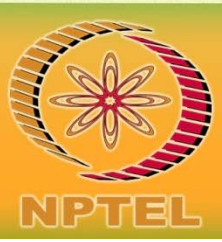


# Code Inspection

- In contrast to code walk through,
  - code inspection aims mainly at discovery of the commonly made errors that usually creep into code due to programmer mistakes and oversights .
  - also aims at checking adherence to coding standards.
- During code inspection:
  - the code is examined for the presence of certain kinds of errors,
  - in contrast to the hand simulation of code execution done in code walk through.

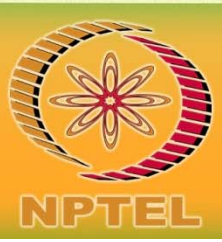


# Benefits of code inspection

- Finds the commonly made errors
- Programmer receives feedback on
  - programming style,
  - choice of algorithm, and
  - programming techniques

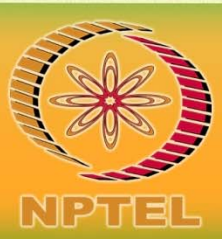


# Code Inspection cont...

- For instance, consider:
  - classical error of writing a procedure that modifies a formal parameter
  - while the calling routine calls the procedure with a constant actual parameter.
- It is more likely that such an error will be discovered:
  - by looking for this kind of mistakes in the code,
  - rather than by simply hand simulating execution of the procedure.

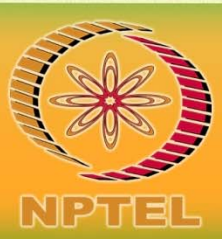


# Code Inspection cont...

- Good software development companies:
  - collect statistics of errors committed by their engineers
  - identify the types of errors most frequently committed.
- A list of common errors:
  - can be used during code inspection to look out for possible errors.

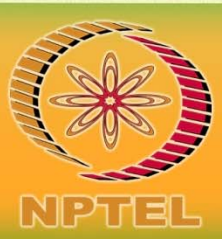


# Examples of some commonly made errors

- Use of uninitialized variables.
- Use of incorrect logical operators or incorrect precedence among operators.
- Non-terminating loops.
- Array indices out of bounds.
- Incompatible assignments.
- Improper storage allocation and deallocation.
- Actual and formal parameter mismatch in procedure calls.
- Jumps into loops
- Improper modification of loop variables, etc.

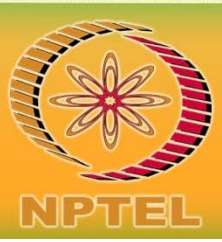


# Cleanroom Technique

- Pioneered at IBM
- The term cleanroom was first coined at IBM by drawing analogy to the semi-conductor fabrication units where the defects are avoided by manufacturing in an ultra-clean atmosphere.
- Relies heavily on walkthroughs, inspection and formal verification for bug removal
- Programmers are not allowed to test any of their code by executing the code other than doing some syntax testing using a compiler
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# Cleanroom Technique cont ...

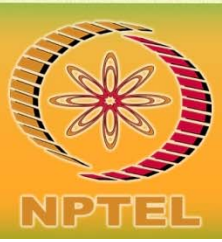
Pros:

- This technique reportedly produces documentation & code that are more reliable and maintainable than other development methods relying heavily on code execution based testing.

Cons:

- The testing effort is increased as walkthroughs, inspection and verification are time consuming for detecting simple errors.
- Some errors might escape during manual inspection. Testing-based error detection techniques may detect these errors.

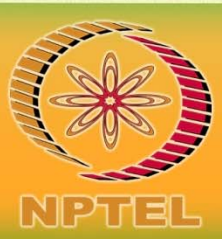


# Summary

- Discussed two important code review techniques such as code walk through and code inspection.
- Presented a list of some commonly made errors which can be detected by code review.
- Discussed briefly Cleanroom Technique .



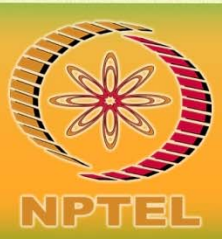



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2. R. Mall, *Fundamentals of Software Engineering*, Fifth Edition, PHI Learning Pvt. Ltd., 2018.








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